Biomarker Interaction Implementation: Architecture Comparison

2026-04-08 09:47 PDT (date of the original render; the creation time was not recorded separately)

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Purpose. This early summary (April 2026) compares the power of the biomarker-by-treatment interaction test under the two data-generating architectures, covariance moderation and mean moderation, across carryover half-lives, and sets out the biological assumptions that would favor each. Its numbers come from a 50-replicate exploratory run and are superseded by later work: see the note below.

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Note added 2026-10-07. The power figures below are exploratory (50 replicates per cell, Monte Carlo standard error up to 0.07) and predate the matched-N re-run of paper 01, which reports Hybrid carryover losses of 6.9% under mean moderation and 15.4% under covariance moderation at $N = 70$ (`analysis/report/NOTATION.md`, Part 2). The mechanistic account in the second section is also superseded: `docs/37` shows, for the paired-difference statistic, that carryover leaves the contrast variance unchanged and acts on power only through the biomarker coupling gap, and `docs/38` argues that the covariance construct keeps the full drug-response variance off drug. The architecture labels A and B used in the original text have been replaced by the canonical names (mean moderation, covariance moderation), and the design labeled "N-of-1" is the Hybrid design.

Notation index and glossary

Table 1: Notation index for the architecture comparison

Symbol	Meaning	Status
N	total number of participants across randomization paths	canonical
c_{bm}	covariance-moderation strength: on-drug correlation of biomarker and BR	canonical
β_{bm}	mean-moderation multiplier	canonical
β_D	treatment main effect	canonical
b, b_1, b_2	standardized biomarker; its values for two participants	canonical
B	biomarker	canonical
BR	biological (drug) response component	canonical
Y	symptom score	canonical
D	binary drug state	canonical
D_{bc}	exposure-decayed drug indicator (code `Dbc`)	canonical
t_{sd}, $t_{1/2}$, λ	time since discontinuation; carryover half-life; decay rate $\ln 2/t_{1/2}$	canonical
σ_{BR}	standard deviation of BR	canonical

Glossary. *Covariance moderation* (formerly Architecture B, data value `mvn`): the interaction is a treatment-state-dependent biomarker-BR correlation. *Mean moderation* (formerly Architecture A, `mean_moderation`): the biomarker shifts the mean drug response. *Hybrid*: the Hendrickson et al. (2020) design (open-label titration, blinded discontinuation, brief crossover), called "N-of-1" in the original tables. *CO*: two-period crossover. *OL+BDC*: open-label titration then blinded discontinuation. *Relative loss*: power at $t_{1/2} = 0$ minus power at $t_{1/2} = 1$, divided by power at $t_{1/2} = 0$.

Empirical Comparison

Identical simulation configuration across both architectures (driver `archive/architecture_comparison/01-co` outside the repository; inspected 2026-10-07):

- Sample size: $N = 70$ in total, allocated across paths by `generateSimulatedResults()`
- Moderation strength: $c_{bm} = \beta_{bm} = 0.45$ (a shared value, per canonical rule 2)
- Data-generating correlations: AR(1) with 0.7 per week within each factor, cross-factor 0.1 at the same visit and 0.05 at different visits
- Analysis model: `nlme::lme` with `corCAR1` and the `bm:Dbc` interaction
- Replicates: 50 per cell

The driver ran two analysis models (half-life ignored, and D_{bc} matched to the data-generating half-life); which one the tables report was not recorded, and the output file (`architecture_comparison_n50.rds`) was not found in the archive, so the tables could not be re-derived. With the current package, every covariance matrix of this configuration is positive definite (smallest eigenvalue 0.043, OL+BDC at $t_{1/2} = 0$; verified 2026-10-07 by `buildSigma()`), so the covariance-moderation cells did not rely on a repaired matrix, provided the April 2026 code built the same matrices (not verified).

Covariance moderation (formerly Architecture B)

Power under carryover half-life variation:

Table 2: Covariance-moderation power by design and carryover half-life (50 replicates per cell)

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
Hybrid	0.82	0.64	0.50
CO	0.40	0.40	0.32
OL+BDC	0.62	0.38	0.24

Power declines substantially with increasing carryover (relative loss from $t_{1/2} = 0$ to 1, computed from the table):

- Hybrid: 39% relative loss (corrected 2026-10-07; the original text gave 68%, which the table does not support)
- CO: 20% relative loss
- OL+BDC: 61% relative loss

Mean moderation (formerly Architecture A)

Power under carryover half-life variation:

Table 3: Mean-moderation power by design and carryover half-life (50 replicates per cell)

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
Hybrid	0.74	0.72	0.68
CO	0.38	0.36	0.34
OL+BDC	0.62	0.58	0.54

Power is largely preserved under carryover:

- Hybrid: 8% relative loss
- CO: 11% relative loss
- OL+BDC: 13% relative loss

At 50 replicates the standard error of a difference between two cells is about 0.09, so only the covariance-moderation declines in Hybrid and OL+BDC are clearly larger than Monte Carlo error; the mean-moderation losses and the CO loss are within it.

Why the Architectures Diverge

Mean moderation: the biomarker shift is confined to on-drug visits

Under mean moderation the package shifts each participant's BR at on-drug visits by $\beta_{bm}\,\sigma_{BR}\,b$ (`R/generateData.R`, default `moderation_scaling = 'constant'`; inspected 2026-10-07). The shift does not depend on the carryover half-life. Carryover acts only on the off-drug BR mean, which carries no biomarker information, so the interaction signal is essentially unchanged; `docs/37` (Section 5.3) derives that, for the paired-difference statistic, power under mean moderation does not depend on $t_{1/2}$ at all.

In the multiplicative form used by paper 01, with treatment main effect β_D, the expected outcome difference between two participants with standardized biomarker values b_1 and b_2 in the same drug state D is

$$E(Y \mid b_1, D) - E(Y \mid b_2, D) = \beta_D\,\beta_{bm}\,(b_1 - b_2)\,D.$$

On drug ($D = 1$) the difference is $\beta_D\beta_{bm}(b_1 - b_2)$ whatever the half-life; off drug it is zero. (Corrected 2026-10-07: the original formula conditioned on $D = 1$

while retaining a factor D, used an undefined β_1, and described the difference as both constant and scaling with carryover.) Under the sign convention of `NOTATION.md`, β_D is negative, so a larger biomarker value lowers the symptom score further on drug.

Covariance moderation: the coupling decays off drug

Under covariance moderation the original account named two mechanisms:

1. **Mean blurring**: the BR means during off-drug periods are inflated by residual carryover, making them resemble on-drug BR means, which was said to reduce the treatment contrast that the analysis model uses to distinguish drug effect from no-drug.
2. **Correlation erosion**: the biomarker-BR correlation during off-drug periods decays exponentially with time since discontinuation (`R/generateData.R`, `buildSigma()`; inspected):

   $$\mathrm{Cor}(B, BR_t) = c_{bm}\, e^{-\lambda t_{sd}} \quad \text{off drug after exposure,}$$

   with c_{bm} on drug and zero before first exposure.

The second rule is a specification choice of the data-generating process, not a structural consequence of it. (Corrected 2026-10-07: the original text called it "not merely a statistical modeling choice" and attributed the decay to a shrinking drug-mediated share of BR variance; the construct holds the BR standard deviation fixed at every visit, so no such share shrinks; `docs/38`, Section 5.)

Later work revises the two-mechanism account (note added 2026-10-07). For the paired-difference statistic, `docs/37` derives that carryover does not change the contrast variance (Proposition 1) and that the mean shift is absorbed by the intercept, so the whole effect of carryover on power runs through the coupling gap between on-drug and off-drug visits (Propositions 2 and 3); mean blurring contributes nothing there. For the `lme` analysis with D_{bc} reported here this was not derived. The losses in the table above are 20-61% (Hybrid 39%), not uniformly 40-60% as the original text stated.

Biological Assumptions

The choice between architectures reflects fundamentally different assumptions about the mechanism by which the biomarker moderates treatment response.

When mean moderation is appropriate

Mean moderation is appropriate when the biomarker governs the **magnitude of the drug's biological effect** for each individual participant. Under this mechanism, the drug effect for a participant is proportional to their biomarker value: higher biomarker predicts larger treatment response.

Example: a pharmacogenetic biomarker that determines hepatic metabolism capacity, where the genotype fixes the size of each participant's response while the drug is present. The interaction is a property of each individual's biology. (Note added 2026-10-07: the original example had high-expression participants metabolizing the drug more rapidly yet responding more strongly; faster metabolism ordinarily lowers exposure and response, so the direction of the example depends on whether the variant affects activation or clearance. The example is illustrative and was not checked against a specific drug.)

When covariance moderation is appropriate

Covariance moderation is appropriate when the biomarker's predictive value emerges from **shared variance** with a drug-responsive physiological component, and this shared variance decays as the drug is eliminated.

Example: a biomarker that reflects drug pathway engagement. The biomarker and drug response are jointly determined by the same underlying pathway state, and their correlation weakens off drug. The interaction is a property of the joint response to the drug, not an inherent property of the individual. (Note added 2026-10-07: in every current data-generating process the biomarker B is a single pre-treatment, participant-level value, so it cannot itself decay during off-drug periods; the original example, an active drug metabolite concentration, describes a time-varying biomarker that the simulation does not represent. Only the coupling $c_{bm}e^{-\lambda t_{sd}}$ decays.)

Selecting the appropriate architecture

For simulation studies of predictive biomarker-moderated treatment effects, the choice should be driven by the clinical hypothesis:

- **Baseline biomarker predicts intrinsic drug responsiveness:** use mean moderation.
- **Biomarker reflects drug pathway engagement:** use covariance moderation.

The power implications are substantial: the choice between architectures determines whether carryover substantially erodes power (covariance moderation) or whether power is largely preserved (mean moderation). This choice must be made based on clinical reasoning, not statistical convenience (argued).

References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020; 2:13. doi:10.3389/fdgth.2020.00013.
2. pmsimstats team. `docs/37-carryover-power-closed-form.md`, `docs/38-pharmacodynamic-carryover-model.md`, `analysis/report/NOTATION.md`.